

Simulation Scenario Report: ED Asthma Exacerbation – Siratech Downtime Protocol

1. Section I: Scenario Demographics

- **Scenario Title:** Emergency Department Operational Readiness – Moderate Asthma Exacerbation during System Downtime.
- **Target Learning Group:**
 - ER Registration & HIM Staff
 - ER Nursing Staff (Triage and Treatment)
 - ER Physicians
 - Laboratory/Phlebotomy & Respiratory Therapy Staff
 - Radiology Technologists
 - Pharmacy Staff
- **Approximate Timing:**
 - **Set-Up:** 30 min
 - **Scenario:** 60 min (to account for manual downtime documentation)
 - **Debriefing:** 20 min

2. Section II: Scenario Developers

- **Stakeholders:**
 - KFSHRC Hospital Operations (ED Leadership)
 - IT/Health IT (Downtime Support)
 - Emergency Nursing & Respiratory Services
 - Health Information Management (HIM)

3. Section III: Curriculum Integration

- **Educational Goal:** To validate the manual downtime workflow, clinical adherence to asthma protocols, and inter-departmental coordination when the Siratech (HIS) is unavailable.
- **CRM Objectives:**
 - **Manual Communication:** Demonstrate effective use of paper orders and verbal communication between ED modules and ancillary services.
 - **Patient Safety during Downtime:** Ensure 100% accuracy in manual patient identification and data tracking without electronic verification.
- **Operational Objectives:**
 - **Downtime Registration:** Validate staff proficiency in activating "downtime processes," including manual MRN assignment and handwritten ID armbands.
 - **Clinical Adherence:** Evaluate the speed of administering nebulization and steroids using downtime medication forms.
 - **Ancillary Coordination:** Confirm manual requisition process for blood work, respiratory swabs, and Chest X-rays.

4. Section IV: Scenario Script

- **Scenario Cast:**
 - **Patient:** Standardized Patient (25-year-old with moderate asthma).
 - **Confederates:** Actual ED staff performing their assigned real-world roles.
- **Required Equipment:**
 - **Downtime Kit:** Manual registration forms, paper lab/rad requisition slips, carbon-copy medication charts.

- **Clinical Tools:** Nebulizer machine, Albuterol/Ipratropium, IV Steroids, Phlebotomy kit, Viral Transport Media (VTM) for respiratory swab.
- **Monitoring:** Continuous Pulse Oximetry and BP monitor.

Scenario Workflow Table:

Step/State	Action / System Trigger	Learner Actions (Downtime Protocol)	Auditor Checklist
Arrival (Visual Triage)	Patient arrives by private car walking to the ER entrance.	• Staff perform immediate Visual Triage. • Direct the patient/relative to the registration. • SARI score • Ask if need assistance.	• Assess speed of initial visual assessment. • Assess the acuricy of SARI score • Assess the encounter
Registration	System Down (Siratech).	• Registrar uses manual downtime log • writes ID armband.	• Verify manual ID accuracy. • Assess the speed time for the registration process
Triage	Comprehensive Assessment.	• Nurse performs manual Triage; • assigns acuity (Level 3). • Bed assignmet to the isolation	• Check manual documentation of vitals.
In emergency room	Physician Assessment.	• Nurse recive the patient (intuduce him/her self) • Nurse notifiy the doctor • Physician intrduce him self • writes Paper Orders for blood work • Medication • Assess the physician encounter	• Ensure "Time Ordered" is recorded.
Treatment	Med Administration.	• Nurse/RT administers nebulization & steroids; Phlebotomist extracts blood.	• Assess "Nurse-to-Patient" med delivery.
Diagnostics	Swab & X-ray.	• Staff perform respiratory swab and transport patient to X-ray with paper slip.	• Verify manual labeling of specimens.
Reassessment	Clinical Review.	• Physician reassesses patient after treatment; documents on paper.	• Evaluate clinical decision-making.
Discharge	Finalizing Care.	• Physician completes manual discharge summary and paper prescriptions.	• Ensure outpatient follow-up is noted.
Pharmacy	Med Collection.	• Patient presents paper script to Pharmacy for discharge medications.	• Assess pharmacist's manual verification.
Post-Discharge		• Patient Experience calls the patient for OPD follow up and patient satisfaction survey	• Assess the Quality of post discharge encounter

5. Section V: Patient Data and Baseline State

- **Clinical Vignette:** "A 25-year-old known asthmatic presents to the ER complaining of shortness of breath and chest tightness for 4 hours. The patient is speaking in partial sentences. Siratech is currently down for unscheduled maintenance."
- **Patient Profile:**
 - **Age:** 25
 - **PMH:** Bronchial Asthma
 - **Allergies:** None
- **Baseline State (Moderate Exacerbation):**
 - **O2 Sat:** 93% on RA (Triggers oxygen/nebulizer protocol)
 - **RR:** 26/min
 - **HR:** 110 bpm
 - **Breath Sounds:** Bilateral expiratory wheezing.

6. Section VI: Supporting Documents & Results

- **Manual Requisitions:**
 - **Lab:** CBC, Electrolytes, Arterial Blood Gas (if indicated).
 - **Radiology:** Portable or Departmental Chest X-ray.
 - **Swab:** Respiratory Viral Panel (RVP).
- **Downtime Log:** Verification that the patient's manual MRN matches all handwritten labels.

7. Section VII: Debriefing Guide

- **Validation Questions:**
 - **Downtime Readiness:** How long did it take to transition from "System Down" awareness to using manual forms?
 - **Communication:** Were the paper orders legible and delivered to the Lab/Radiology without delay?
 - **Patient Safety:** Was the manual ID armband checked before the administration of steroids and nebulization?
 - **Workflow:** Did the lack of Siratech notifications (SMS or digital cues) lead to any "dead zones" in the patient's care?